The complement of K_0 is not computably enumerable

The Open Logic Project contributors

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Editorial note. All delivered mathematical text is the author's own, copied from the pinned Open Logic Project unit file sources/olp-computability-computability-theory/pinned-original-source.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The counted proof is the author's complete argument: the corollary is short by nature: the author's argument is closed (no omitted case), resting on the two cited facts which are both registered/stubbed

Editorial note. The label cor:comp-k names the counted statement, and the proof environment immediately following it in the source is delivered verbatim as its proof.

Editorial note. The counted statement is carried by the author's own cor environment in the unit source.

Allowed standard prerequisites (editorial).

- Characterisation of computability by c.e. of a set and its complement. Same unit, \ollabel{thm:ce-comp} (source lines 678-682). If the complement of K_0 were c.e., thm:ce-comp would force K_0 computable.
- K_0 is c.e. but not computable. Same unit, \ollabel{thm:K-0} (source lines 1663-1666). Supplies both facts used: K_0 c.e., and K_0 not computable (the contradiction horn).

Source reference. Original theorem thm-ce-comp: sources/olp-computability-computability-theory/pinned-original-source.tex, source lines 678–682. Prerequisite pointer invoked in the counted proof; it names a standard fact of the same unit that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original theorem thm-K-0: sources/olp-computability-computability-theory/pinned-original-source.tex, source lines 1663–1666. Prerequisite pointer invoked in the counted proof; it names a standard fact of the same unit that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Corollary 0.1. *$\overline{K_0}$ is not computably enumerable.*

Proof. We know that K_0 is computably enumerable, but not computable. If $\overline{K_0}$ were computably enumerable, then K_0 would be computable by thm-ce-comp, contradicting thm-K-0. □